

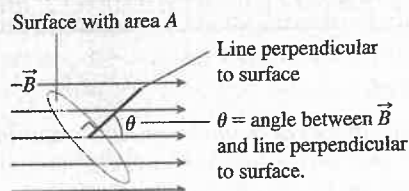
CHAPTER 19 SUMMARY

Induction and Faraday's Law

(Section 19.1) In electromagnetic induction, a **changing magnetic flux** induces an emf, as described by **Faraday's law**. The emf gives rise to a current if a complete circuit is present. **Lenz's law** states that the induced current opposes the change in magnetic flux.

Magnetic flux: $\Phi = BA \cos \theta$

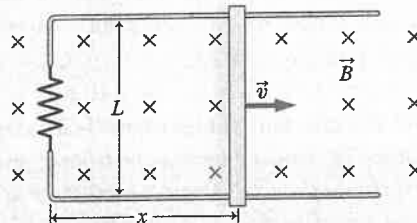
Faraday's law of induction: $\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$ for an N -turn coil



Motional emf

(Section 19.2) **Motional emf** is the emf induced when there is relative motion between a conductor and a magnetic field. If the conductor is a solid piece, induced **eddy currents** develop and dissipate energy. This effect is the basis of magnetic braking and other applications.

Current induced by motional emf in closed loop: $I = \frac{|\mathcal{E}|}{R} = \frac{BLv}{R}$

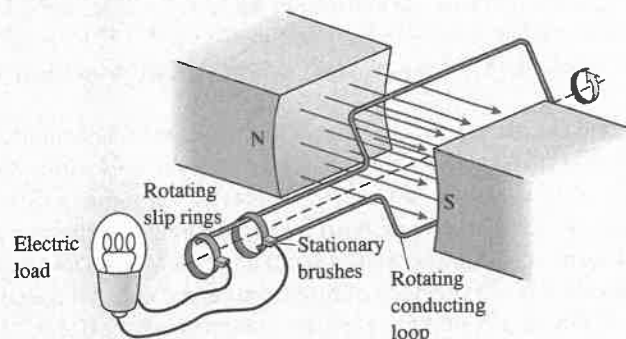


Generators and Transformers

(Section 19.3) An electric generator converts mechanical energy to electrical energy. The **rotating-coil generator** consists of conducting coils rotating in a magnetic field; it produces alternating current at frequency equal to its rotation rate. **Transformers** use electromagnetic induction to change emf in AC circuits.

emf induced in a rotating N -turn coil: $\mathcal{E} = NBA\omega \sin(\omega t)$

Transformer emf: $\mathcal{E}_s = \mathcal{E}_p \frac{N_s}{N_p}$



Inductance

(Section 19.4) Through **inductance**, a changing current in one circuit induces current in another circuit. An **inductor** is a coil designed for its inductive properties. The reciprocal effect between two circuits is known as **mutual inductance**. A single coil of wire can induce a current in itself through **self-inductance**. Inductors store energy in the magnetic field. When an inductor is placed in a circuit with a resistor (an **RL circuit**), inductance delays the growth of current in the circuit. When an inductor is placed in series with a capacitor (an **LC circuit**), the charge oscillates.

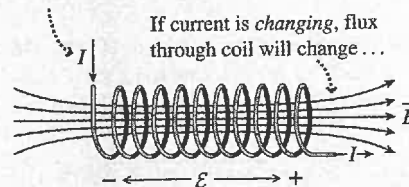
Definition of inductance: $L = \frac{N\Phi}{I}$

Energy stored in an inductor: $U = \frac{1}{2} LI^2$

Energy density in a magnetic field: $u_B = \frac{B^2}{2\mu_0}$

Charge oscillation in LC circuit: $Q = Q_0 \cos(\omega t)$

Current in wire creates magnetic flux through coil.



If current is *changing*, flux through coil will change ...
... inducing an emf in the coil that *opposes* the change in current.

AC Circuits

(Section 19.5) The AC current to a capacitor leads the capacitor's potential difference by 90° . The AC current to an inductor lags the inductor's potential difference by 90° . Capacitors and inductors have **reactance**, which acts somewhat like resistance but also introduces a phase difference between current and potential difference. Phasors allow graphical analysis of AC circuits.

rms current: $I_{\text{rms}} = I_{\text{max}}/\sqrt{2}$

Capacitive reactance: $X_C = 1/\omega C$

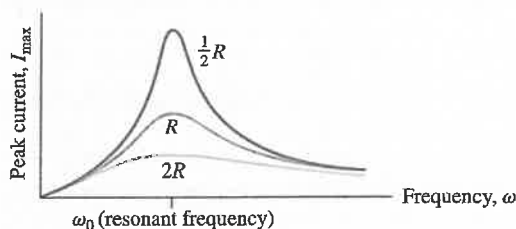
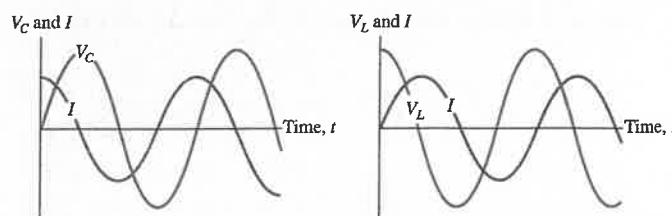
Inductive reactance: $X_L = \omega L$

RLC Circuits and Resonance

(Section 19.6) An *RLC* series circuit has an **impedance** Z , analogous to the resistance of a DC circuit. The **phase angle** ϕ tells how much the current is ahead of or behind the emf. **Resonance** occurs when $X_C = X_L$ and is analogous to resonance in a mechanical oscillator.

Peak current in *RLC* series circuit: $I_{\text{max}} = \frac{\mathcal{E}_{\text{max}}}{\sqrt{R^2 + (X_L - X_C)^2}}$

Resonant frequency: $\omega_0 = \sqrt{\frac{1}{LC}}$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled B10 are of biological or medical interest.

Conceptual Questions

- Show that with SI units on the right side of Faraday's law (Equation 19.2), the emf comes out in volts.
- Explain why magnetic flux decreases as the angle θ increases from 0 to 90° .
- Is the magnetic flux through a horizontal loop larger at the equator or the North Pole?
- Explain why the magnetic brake shown in Figure 19.10 works equally well when the metal plate is moving into or out of the magnetic field but doesn't work when the plate is entirely within the uniform magnetic field.
- In the dropping-magnet demonstration (Conceptual Example 19.4), the magnet has very little kinetic energy when it exits the tube. What happened to most of the gravitational potential energy it had at the top?
- Suppose in Figure 19.7 that the magnetic field is out of the page instead of in. What are (a) the direction of the induced current, (b) the induced emf, and (c) the direction of the force on the sliding bar?
- Suppose the bar in Figure 19.7 moves leftward instead of rightward. What are (a) the direction of the induced current, (b) the induced emf, and (c) the direction of the force on the sliding bar?
- In an *RL* circuit, is there more energy in the inductor just after the switch is closed or much later?
- How is an *LC* circuit like a simple harmonic oscillator?
- How is an *RLC* circuit like a damped harmonic oscillator?
- Suppose you know the resistance, inductive reactance, and capacitive reactance for a series *RLC* circuit. Is the total impedance for the circuit equal to the sum of these three quantities? Why or why not?

- For which of the following AC circuits does impedance depend on frequency? A circuit with (a) only a resistor; (b) a resistor and inductor; (c) a resistor and capacitor.
- In a series *RLC* circuit, is energy dissipated in the inductor or capacitor? Explain.
- A series *RLC* circuit has $X_C > X_L$. To bring the circuit into resonance, should you increase or decrease the capacitance?
- An *RLC* circuit has power factor 0.90. From this information, can you tell whether the current leads or lags the emf?

Multiple-Choice Problems

- A 2.3-T magnetic field lies perpendicular to a 35-cm-diameter circular wire loop. The magnetic flux through the loop is (a) $0.07 \text{ T} \cdot \text{m}^2$; (b) $0.14 \text{ T} \cdot \text{m}^2$; (c) $0.22 \text{ T} \cdot \text{m}^2$; (d) $0.28 \text{ T} \cdot \text{m}^2$.
- A 1.9-T magnetic field makes a 60° angle with a square loop 50 cm on a side. The magnetic flux through the loop is (a) $0.24 \text{ T} \cdot \text{m}^2$; (b) $0.41 \text{ T} \cdot \text{m}^2$; (c) $0.48 \text{ T} \cdot \text{m}^2$; (d) $0.75 \text{ T} \cdot \text{m}^2$.
- A 130-turn circular coil has diameter 5.2 cm. A magnetic field perpendicular to the coil is changing at 0.75 T/s . The induced emf in the coil is (a) 0.07 V; (b) 0.14 V; (c) 0.21 V; (d) 0.28 V.
- A 50-turn circular coil has diameter 6.2 cm and resistance 0.75Ω . A magnetic field perpendicular to the coil is changing at 0.50 T/s . The induced current in the coil is (a) 10 mA; (b) 30 mA; (c) 100 mA; (d) 300 mA.
- For the apparatus shown in Figure 19.7, the magnetic field strength is 4.5 T and the rail separation is $L = 25 \text{ cm}$. What's the induced emf when the bar moves at 35 cm/s ? (a) 0.39 V; (b) 0.45 V; (c) 0.53 V; (d) 0.65 V.

21. A generator has a coil with area 0.12 m^2 , rotating at 60 Hz in a 0.47-T magnetic field. If the generator's peak emf is 340 V , the number of turns in the coil is (a) 16; (b) 32; (c) 50; (d) 100.
22. The primary coil in a transformer has 100 turns and a 120-V emf. If you want a secondary emf of 240 V , the number of turns in the secondary coil should be (a) 25; (b) 50; (c) 200; (d) 400.
23. You have a 320-mH inductor. Over what time must you increase the inductor current by 2.0 A in order to induce a 5.0-V emf? (a) 0.13 s ; (b) 0.26 s ; (c) 0.39 s ; (d) 0.52 s .
24. A radio receiver has a $25\text{-}\mu\text{H}$ inductance. The capacitance required to tune in a 1470-kHz radio station is (a) 0.47 nF ; (b) 1.88 nF ; (c) 3.76 nF ; (d) 18.8 nF .
25. An AC power source produces a 170-V peak emf at 60 Hz . What's the average power dissipated in a $480\text{-}\Omega$ resistor connected across this source? (a) 75 W ; (b) 60 W ; (c) 42 W ; (d) 30 W .
26. A variable capacitor is connected across a variable-frequency AC source. The combination that gives the highest current is: (a) $23 \text{ }\mu\text{F}$, 60 Hz ; (b) $33 \text{ }\mu\text{F}$, 55 Hz ; (c) $30 \text{ }\mu\text{F}$, 70 Hz ; (d) $25 \text{ }\mu\text{F}$, 50 Hz .

Problems

Section 19.1 Induction and Faraday's Law

27. ■ A 1.5-T magnetic field lies perpendicular to a 25-cm -diameter circular wire loop. What's the magnetic flux through the loop?
28. ■ A 2.9-T magnetic field makes a 40° angle with a square loop 25 cm on a side. What's the magnetic flux through the loop?
29. ■■ At one location, Earth's magnetic field has a magnitude of $5.4 \times 10^{-5} \text{ T}$ with an inclination of 72° to the horizontal. Find the magnetic flux through a horizontal rectangular roof measuring 35 m by 20 m .
30. ■■ A circular wire loop of wire has radius 2.0 cm and resistance $0.050 \text{ }\Omega$. A magnetic field is perpendicular to the loop. At what rate must the field change in order to induce a current of 1.0 mA in the loop?
31. ■■ A 150-turn circular coil has diameter 5.25 cm and resistance $1.30 \text{ }\Omega$. A magnetic field perpendicular to the coil is changing at 1.15 T/s . Find the induced current in the coil.
32. ■■ A circular loop of wire with area 0.015 m^2 lies in the $x\text{-}y$ plane. Initially there's a magnetic field of 4.0 T in the $-z$ -direction. The field remains constant for 10 s , then decreases gradually to zero in 10 s , and then remains zero for 10 s . Find the magnitude and direction of the induced emf in the loop for each of the three 10-s intervals.
33. ■■ A square wire loop 15 cm on a side lies in the $x\text{-}y$ plane. Its resistance is $0.55 \text{ }\Omega$. A magnetic field points in the $+z$ -direction and increases at 15 mT/s . (a) What's the direction of the induced current in the loop? (b) Find the magnitudes of the induced emf and induced current in the loop.
34. ■■ A 75-turn circular coil has diameter 4.0 cm and resistance $2.6 \text{ }\Omega$. This coil is placed inside a solenoid, with coil and solenoid axes aligned. The solenoid has 5000 turns of wire and is 24 cm long. If the solenoid current increases steadily from 0 to 10 A in 2.5 s , find the induced emf and induced current in the coil.
35. ■■ A circular wire coil with resistance $1.4 \text{ }\Omega$ and area $5.0 \times 10^{-3} \text{ m}^2$ lies perpendicular to a magnetic field that's increasing at 2.0 T/s . If the induced current is 250 mA , how many turns are in the coil?

Section 19.2 Motional emf

36. ■ In Figure 19.7, take $B = 3.5 \text{ T}$ and $L = 12 \text{ cm}$. Find the induced emf when the bar moves at 0.25 m/s .
37. ■ In Figure 19.7, take $B = 1.8 \text{ T}$, $R = 2.3 \text{ }\Omega$, and $L = 12 \text{ cm}$. Suppose the bar moves at 0.80 m/s to the left. (a) Find the magnitude and direction of the induced current. (b) At what rate is electric power generated?
38. ■■ Suppose the rectangular wire loop with a resistance of $3.5 \text{ }\Omega$ enters the magnetic field as shown in Figure P19.38, moving to the right at 25 cm/s . (a) What are the magnitude and direction of the induced current? (b) Later the loop is fully in the field, still moving at 25 cm/s . What are the magnitude and direction of the induced current? (c) Eventually the loop begins to emerge from the field, still at 25 cm/s . What are the magnitude and direction of the induced current?

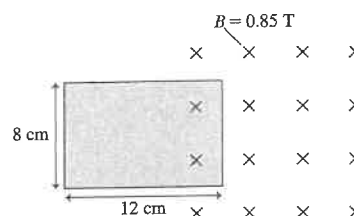


FIGURE P19.38

39. ■■ The apparatus in Figure 19.7 is rotated 90° , so the bar falls vertically while maintaining electrical contact with the rails. The rail spacing is $L = 30 \text{ cm}$, and the field strength is $B = 1.2 \text{ T}$. If the bar drops from rest, what's the magnitude of the induced emf after it's been falling for (a) 0.5 s and (b) 1.0 s ? (You may assume that the magnetic force on the falling bar is much less than the bar's weight.)
40. ■■ In the rail gun of Figure 19.12, the field strength is 3.0 T and the rail spacing is 50 cm . A discharging capacitor supplies a current of 100 kA . (a) What's the acceleration of the 0.25-kg bar? (b) What's the bar speed at the end of the 2.75-m track?
41. ■■ A horizontal metal bar of length L falls vertically through a horizontal magnetic field of strength B . (a) Show that if the bar falls with speed v , the induced emf between its ends is $\mathcal{E} = BLv$. (b) Suppose $B = 0.25 \text{ T}$ and $L = 0.50 \text{ m}$. What's the induced emf 1.0 s after the bar is dropped from rest?
42. ■■ An airplane flies due north in a region where Earth's magnetic field has magnitude $5.3 \times 10^{-5} \text{ T}$, declination 0° , and inclination 75° . What's the induced emf between the wingtips, 35 m apart, when the plane flies at 220 m/s ? *Hint:* See the preceding problem.

Section 19.3 Generators and Transformers

43. ■ (a) A 20-turn generator coil with area 0.016 m^2 rotates at 50 Hz in a 0.75-T magnetic field. Find the peak induced emf. (b) Graph the induced emf as a function of time from $t = 0$ to $t = 40 \text{ ms}$.
44. ■ A 10-turn generator coil with area 0.150 m^2 rotates in a 1.25-T magnetic field. At what frequency should the coil rotate to produce a peak emf of 150 V ?
45. ■■ A generator has a 150-turn square coil, 5 cm on a side. Find the peak emf in this coil when it's rotating at 100 Hz (a) in Earth's field, $B = 5 \times 10^{-5} \text{ T}$, and (b) in a strong 6.4-T field.
46. ■■ Figure P19.46 shows the induced emf in a 5-turn square coil, 8 cm on a side, rotating in a uniform magnetic field. Find (a) the rotation frequency, (b) the angular velocity, and (c) the magnetic field strength.

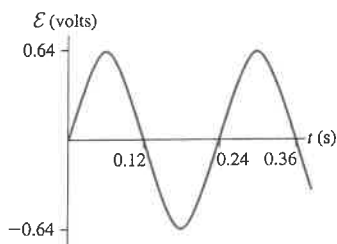


FIGURE P19.46

47. ■ (a) A 120-V emf is across a transformer's 200-turn primary coil. How many turns should the secondary have in order to produce a 30-V emf? (b) A 240-V emf is across a transformer's 140-turn primary coil. What's the emf in the 250-turn secondary coil?
48. ■■ A home receives 1.5 kW of electric power from a transformer on a nearby pole. The line from the substation to the transformer is at 10 kV and has $0.13\ \Omega$ resistance. What fraction of the power is lost in this transmission line?
49. ■■ A typical coal-fired power plant produces 1000 MW of electrical energy. Seven percent of the power is lost in a 500-kV transmission line. (a) What's the line's resistance? (Note: This is actually the equivalent resistance of all the parallel lines in the network, so your answer should be quite small.) (b) What would be the power loss if the line were at 100 kV?

Section 19.4 Inductance

50. ■ A 240-turn coil carries 350 mA. If the magnetic flux through the coil due to this current is 0.75 Wb, what's the coil's inductance?
51. ■ You have a 120-mH inductor. Over what time must you increase the current through the coil by 3.0 A to induce a 6.0-V emf?
52. ■■ A solenoid has diameter 2.5 cm, length 30 cm, and inductance 1.34 mH. (a) How many turns of wire does it have? (b) What's the energy stored in the solenoid when it carries a 6.0-A current?
53. ■■ A solenoid has the following dimensions: diameter = 5.0 cm, $n = 45$ turns/cm, length = 23 cm. (a) What's its inductance? (b) Find the current when the solenoid stores 0.50 J of energy.
54. ■ Find the inductance required to store 1.0 J of energy with current of 10 A through the inductor.
55. ■■ A 48-V battery is in series with a switch, a 0.50-H inductor, and a $10\text{-}\Omega$ resistor. (a) What's the maximum current in this circuit? (b) What's the current at $t = 0.10$ s after the switch is closed? (c) When does the current reach half the maximum value? (d) What's the current at time $t = 2\tau$?
56. ■■ A 12.0-V battery is in series with a switch, a 0.25-H inductor, and a $20\text{-}\Omega$ resistor. (a) What's the time constant τ ? (b) Find the potential differences across the inductor and resistor at $t = \tau$ and $t = 2\tau$.
57. ■■ A 10.0-mH inductor is in series with a capacitor that can be varied from 200 nF to 400 nF. What's the range of possible oscillation frequencies?
58. ■ You have a variable capacitor ranging from 5.9 pF to 8.8 pF. What inductor should be used with this capacitor in an FM radio receiver covering the entire FM band, 88 MHz to 108 MHz?
59. ■■ A charged, 250- μ F capacitor is connected across a 450-mH inductor. (a) What's the oscillation period? (b) How should you change the capacitance in order to double the period?
60. ■■■ A 4700- μ F capacitor is charged to 9.0 V and then connected across a 1.50-H inductor. (a) What's the energy in this circuit? (b) What's the maximum current? (c) At what time after the circuit is connected are there equal energies in both components?

61. ■■ A 500- μ F capacitor is connected across a 1.25-H inductor. At a certain time, the charge on the capacitor is zero and the current is 0.342 A. (a) How much later will the capacitor charge reach its peak? (b) What's the total energy in the circuit? (c) What is the peak charge on the capacitor?
62. ■■■ (a) Consider the circuit shown in Figure P19.62. Initially there's no current in the circuit. Find the current in each resistor immediately after the switch is closed and a long time later. (b) After the switch has been closed for a long time, it's opened. Find the current in each resistor immediately after it opens.

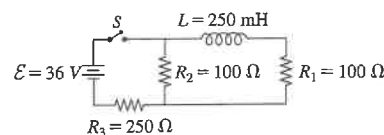


FIGURE P19.62

Section 19.5 AC Circuits

63. ■ An AC power supply operates with peak emf 340 V and frequency 120 Hz. A 3.50-k Ω resistor is connected across the supply. Find (a) the rms current and (b) the average power dissipated in the resistor. (c) Graph the potential difference across and current through the resistor, both as functions of time.
64. ■■ A capacitor is connected to a 200-V (peak), 50-Hz power supply. If the maximum current is 400 mA, what is the capacitance?
65. ■■ You wish to design an AC power supply that can drive 2.0 A rms current through a 140- Ω load. (a) What should be the rms value of the power supply's emf? (b) At what rate does this power supply deliver energy?
66. ■■ When a 750- Ω resistor is connected across an AC power supply, the resistor dissipates energy at a rate of 12.5 W. (a) Find the rms current and maximum current through the resistor. (b) What is the rms value of the power supply's emf?
67. ■■■ An AC power supply delivers 120 V rms. The time interval between peak emf in one direction and the opposite direction is 10.2 ms. Find (a) the peak emf and (b) the frequency. (c) Write an expression for the power supply's emf as a function of time, including all appropriate numerical values.
68. ■■ A resistor connected across an AC power supply has a current given by $I = (1.20\text{ A}) \cos(300t)$ when connected to a power supply with emf 100 V rms. Find (a) the rms current, (b) the resistance, and (c) the average power delivered to the resistor.
69. ■ (a) At what frequency does a 100- μ F capacitor have 50- Ω reactance? What would be its reactance if the frequency were tripled? (b) At what frequency does a 100-mH inductor have 50- Ω reactance? What would be the reactance of this inductor if the frequency were tripled?
70. ■■ A 240-V (rms), 50-Hz power supply is connected to a 4700- μ F capacitor. Find (a) the capacitor's reactance and (b) the peak current. (c) What's the peak current if the power supply's frequency is changed to 60 Hz?

Section 19.6 RLC Circuits and Resonance

71. ■■ A solenoid has resistance $13.5\ \Omega$ and inductance 410 mH. When it's connected across a 50-Hz AC power supply, what's the circuit's impedance?
72. ■■ An RC series circuit has $R = 75.0\ \Omega$ and $C = 580\text{ nF}$. The series combination is connected across a 200-V (rms), 60-Hz

- power supply. Find (a) the phase angle, (b) the impedance of the RC combination, and (c) the peak current.
73. ■■ An RC series circuit has $R = 500\ \Omega$ and $C = 200\ \mu\text{F}$. The series combination is connected across a 120-V (rms) power supply. Find the peak current when the frequency is (a) 60 Hz and (b) 120 Hz.
74. ■■ An RL series circuit has $R = 75.0\ \Omega$ and $L = 1.90\ \text{mH}$. The series combination is connected across a 150-V (rms), 60-Hz power supply. Find (a) the phase angle, (b) the impedance of the RL combination, and (c) the peak current.
75. ■■■ An RLC series circuit has $R = 1.35\ \text{k}\Omega$, $L = 225\ \text{mH}$, and $C = 2.50\ \mu\text{F}$ and is connected across a 60-Hz power supply. Find (a) the reactances of the capacitor and inductor and (b) the impedance of the circuit. (c) To what value should you change the capacitance in order to bring the circuit into resonance?
76. ■■■ A 120-V (rms), 60-Hz power supply is connected to an RLC series circuit with $R = 150\ \Omega$, $L = 1.20\ \text{mH}$, and $C = 33.5\ \mu\text{F}$. Find (a) the reactance of the capacitor and inductor, (b) the circuit impedance, and (c) the peak current.
77. ■■ For the circuit in the preceding problem, find (a) the phase angle, (b) the power factor, and (c) the average power consumption.
78. ■■ An RLC series circuit is in resonance at 60 Hz. If you triple both the inductance and capacitance, what's the new resonance frequency?
79. ■■ An RLC series circuit has $R = 920\ \Omega$, $L = 15.0\ \text{mH}$, and $C = 250\ \mu\text{F}$. (a) What's its resonance frequency? (b) If the circuit is connected to a 100-V rms power source at the resonance frequency, what's the average power consumption?
80. ■■■ An RLC series circuit has $R = 1.10\ \text{k}\Omega$, $L = 190\ \text{mH}$, and $C = 4.50\ \mu\text{F}$. (a) What frequencies will give a power factor of (a) 0.5; (b) 0.75; (c) 1.0?

General Problems

81. ■■ A car alternator produces about 14 V peak output to charge the car's 12-V battery. If an alternator coil is 15 cm in diameter and is spinning at 1200 revolutions per minute in a 0.15-T magnetic field, how many turns must it have to produce a 14-V peak output?
82. ■■ A square loop 3.0 m on a side is perpendicular to a uniform 2.0-T magnetic field as shown in Figure GP19.82. A 6.0-V light-bulb is in series with the loop. The magnetic field is reduced steadily to zero during a time interval Δt . (a) Find Δt such that the bulb will shine at full brightness during this time. (b) In which direction does the loop current flow?

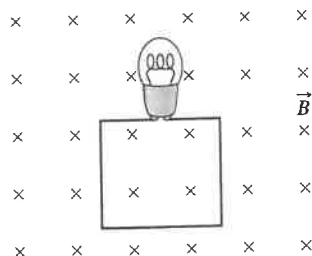


FIGURE GP19.82

83. ■■ Opening switches in highly inductive circuits can be dangerous, even with low-voltage power sources. To see why, consider a 6-V battery supplying 3.0 A to an electromagnet with inductance 2.5 H. A switch in this circuit opens and the current drops to zero in a mere 6.8 ms. What's the emf induced in the electromagnet during this time?
84. B10 ■■ **MRI scanner.** An MRI scanner uses a large solenoid with inductance 0.53 H, carrying 2.4 kA of current. (a) What's the energy stored in the solenoid? (b) The solenoid coils are normally superconducting (zero resistance), but if superconductivity fails they revert to a resistance of 0.31 m Ω . (b) Given that the current can't change instantaneously, find the power dissipation in the solenoid immediately after a sudden loss of superconductivity.
85. ■■■ A 3.2-cm-diameter circular coil with 30 turns of wire has resistance 1.3 Ω . This coil is placed inside a solenoid, with coil and solenoid axes aligned. The solenoid has 3000 turns and is 25 cm long. (a) At what rate must the current in the solenoid be increased to induce a 1.0-mA current in the coil? (b) Is the induced current in the same direction or opposite the solenoid current?
86. ■■ An electric toothbrush rests on a stand containing a 50-turn coil connected across the 120-V rms AC power line. Inside the brush handle is another coil whose induced current charges a 6-V battery. How many turns should this coil contain if its peak emf is to be 7.1 V?
87. ■■■ Earth's magnetic field has an approximate value of 50 μT . (a) Estimate the total magnetic energy contained in the first 100 km above Earth's surface. (b) If we could tap this energy, how long would it supply humankind's power demand of 15 TW?

Answers to Chapter Questions

Answer to Chapter-Opening Question

The power factor $\cos \phi$ describes the phase difference between current and potential difference in AC circuits. A heavy air conditioning load during a hot August week in 2003 lowered the power factor, resulting in higher-than-normal currents to deliver a given power. High ambient temperatures and the high current caused power lines to overheat, sag, and short-circuit to trees. Disruptions spread, taking more than 500 generators offline, including 22 nuclear power plants.

Answers to GOT IT? Questions

- Section 19.1 (c) > (a) > (b) > (d)
 Section 19.2 (a) Clockwise
 Section 19.3 (b) 50
 Section 19.4 (e) 710 kHz
 Section 19.5 (a) > (c) > (b)
 Section 19.6 (a) 0.25